Moment Generating function :-

Moment generating function of a random variable x (about origin) is given by

$$M_x(t) = E(e^{tx})$$

$$= \int_{-\infty}^{\infty} e^{tx} \cdot f(x)\, dx, \quad \text{for cont. prob distribution}$$

$$= \sum_x e^{tx} p(x), \quad \text{for discrete prob distribution.}$$

t is real parameter. The integration or summation being extended to the entire range of x.

$$M_x(t) = E(e^{tx})$$

$$= E\left(1 + tx + \frac{t^2x^2}{2!} + \frac{t^3x^3}{3!} + \cdots + \frac{t^r x^r}{r!} + \cdots\right)$$

$$= E(1) + tE(x) + \frac{t^2}{2!}E(x^2) + \frac{t^3}{3!}E(x^3) + \cdots$$

$$+ \cdots + \frac{t^r}{r!}E(x^r) + \cdots$$

as $E(C) = C$

$$M_x(t) = 1 + t\mu_1' + \frac{t^2}{2!}\mu_2' + \cdots + \frac{t^r}{r!}\mu_r' + \cdots$$

$$= \sum_{r=0}^{\infty} \frac{t^r}{r!}\mu_r'$$

where
$$\mu_r' = E(X^r) = \begin{cases} \sum_x x^r p(x) & \text{for dis. r.v} \\ \int x^r f(x)\,dx & \text{cont. r.v.} \end{cases}$$

μ_r' is r^{th} moment of X about origin.

ie its clear that coefficient of $\frac{t^r}{r!}$ in $M_X(t)$ gives μ_r' (about origin).

Since $M_X(t)$ generates moments, it is known as moment generating function.

Now
$$\frac{d}{dt} M_X(t)\Big|_{t=0} = \mu_1'$$

$$\frac{d^2}{dt^2} M_X(t)\Big|_{t=0} = \mu_2'$$

$$\vdots$$

In gene.
$$\frac{d^r}{dt^r} M_X(t)\Big|_{t=0} = \mu_r'$$

Moment Generating function X about the point X = a is defined as

$$M_X(t) \text{ (about } X=a) = E(e^{t(X-a)})$$

$$= E\left(1 + t(X-a) + \frac{t^2}{2!}(X-a)^2 + \cdots + \frac{t^r}{r!}(X-a)^r + \cdots\right)$$

$$= 1 + t\mu_1' + \frac{t^2}{2!}\mu_2' + \cdots + \frac{t^r}{r!}\mu_r' + \cdots$$

where $\mu_r' = E[(X-a)^r]$ is r^{th} moment about X = a

Q:- Find m.g.f of random variable with prob. law $P(X=x) = q^{x-1}p$, $x = 1, 2, 3$
find mean and variance.

Soln:- $M_X(t) = E(e^{tX})$

$$= \sum_x e^{tx} p(x)$$

$$= \sum_{x=1}^{\infty} e^{tx} q^{x-1} p$$

$$= q^{-1} p \sum_{x=1}^{\infty} e^{tx} q^x$$

$$= q^{-1} p \sum_{x=1}^{\infty} (e^t q)^x$$

$$= q^{-1} p \left[e^t q + (e^t q)^2 + (e^t q)^3 + \cdots \infty \right]$$

$$= q^{-1} p \, e^t q \left(1 + e^t q + (e^t q)^2 + \cdots \infty \right)$$

$$= p e^t \left(\frac{1}{1 - q e^t} \right)$$

$$\boxed{M_X(t) = \frac{p e^t}{1 - q e^t}}$$

Mean = first moment about origin

$$\mu_1' = \frac{d M_X(t)}{dt}\Big|_{t=0} = \frac{(1 - q e^t) p e^t - (-q e^t) p e^t}{(1 - q e^t)^2}\Big|_{t=0}$$

$$= \frac{p e^t}{(1 - q e^t)^2}\Big|_{t=0} = \frac{p}{(1-q)^2}$$

as $p+q=1$

$$M_x'(0) = \mu_1' = \frac{1}{p}$$

Variance :-

$$\text{variance} = \mu_2' - \mu_1'^2$$

$$\mu_2' = \left.\frac{d^2}{dt^2}(M_x(t))\right|_{t=0}$$

$$= \left.\frac{pe^t(1+qe^t)}{(1-qe^t)^3}\right|_{t=0}$$

$$\mu_2' = \frac{p(1+q)}{(1-q)^3} = \frac{1+q}{p^2}$$

$$\text{variance} = \mu_2' - \mu_1^2$$

$$= \frac{1}{p^2} + \frac{q}{p^2} - \frac{1}{p^2} = \frac{q}{p^2}$$

Q:- Find M.G.F of R.V whose moments are

$$\mu_r' = (r+1)!\, 2^r$$

Solⁿ:

$$M_x(t) = \sum_{r=0}^{\infty} \frac{t^r}{r!} \mu_r'$$

$$= \sum_{r=0}^{\infty} \frac{t^r}{r!} (r+1)!\, 2^r$$

$$M_x(t) = \sum_{r=0}^{\infty} (2t)^r \frac{(r+1)!}{r!} = \sum_{r=0}^{\infty} (2t)^r r$$

$$= 1 + 2(2t) + 3(2t)^2 + \cdots$$

$$= \frac{1}{(1-2t)^2}$$

Q:- Spse that you have a fair 4 sided die and let X be the random variable reprsenting the value of the no. rolled.

(1) write m.g.f of X.
(2) find m.g.f to compute first and second moment of X.

Sol^n:- Let $X = 1, 2, 3, 4$

$P(X=1) = \frac{1}{4}$, $P(X=2) = \frac{1}{4}$, $P(X=3) = \frac{1}{4}$

$P(X=4) = \frac{1}{4}$

$$M_X(t) = E(e^{tX})$$

$$= \sum_{x=1}^{4} e^{tx} p(x)$$

$$= e^t p(1) + e^{2t} p(2) + e^{3t} p(3) + e^{4t} p(4)$$

$$M_X(t) = \frac{e^t + e^{2t} + e^{3t} + e^{4t}}{4}$$

(2) $$M_X'(t) = \frac{e^t + 2e^{2t} + 3e^{3t} + 4e^{4t}}{4}$$

$$E(X) = \mu_1' = M_X'(0) = 5/2$$

$$E(X^2) = M_X''(t)\Big|_{t=0} = \frac{15}{2}$$